A REVIEW AID FOR GERMAN CREDIT APPLICANTS

# Credit Risk Auditor

Score, explanation, lending clauses, human review

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CONTEXT

# The problem

### Flag vs. Reason

Model flags lack actionable justification.

### Weight vs. Policy

Positive weights may violate adverse rules.

### Label vs. Score

Historical outcomes differ from risk scores.

ARCHITECTURE

# What the system does

1

**XGBoost**

Scores predicted default risk.

2

**SHAP**

Maps path from average risk.

3

**Clauses**

Retrieves official EBA guidelines.

4

**Draft**

Writes optional reviewer summary.

5

**Human**

Makes final lending decision.

DATASET

# Data

**1,000**

**Applicants, 20 Attributes**

UCI / Statlog German Credit

**Binary classes:** Class 1 good, class 2 bad credit.

**Target encoding:** Bad credit modeled as positive class.

**Checking status:** Balance below 0 DM.

CLASSIFIER

# Model

**78%**
Held-out accuracy

**0.815**
ROC-AUC score

**50%**
Reviewer flag cutoff

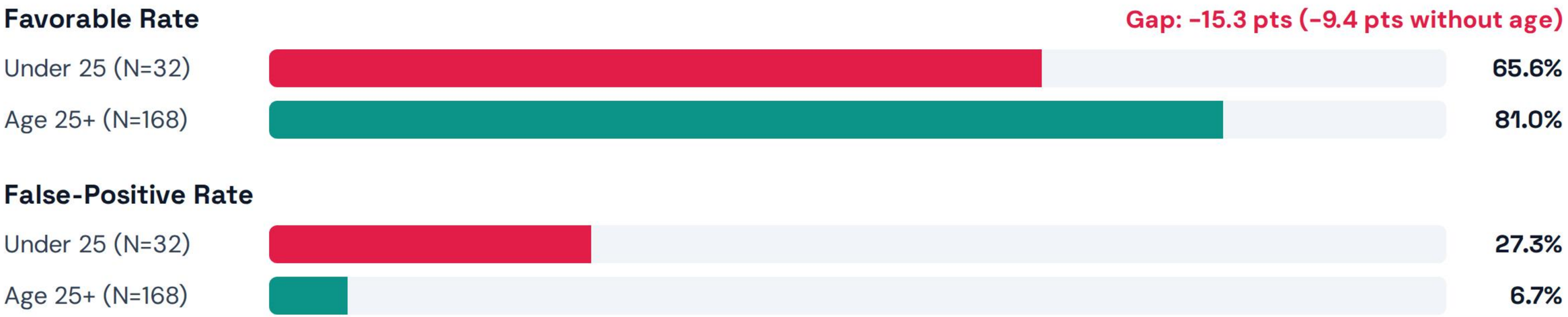
**160 KB**
Shared artifact size

**Architecture:** XGBoost, depth 4, 80/20 split, seed 42.

**Reproducibility:** Identical scores across all machines.

AUDIT GROUP: UNDER 25

# Fairness




Sex is coupled with marital status; non-foreign test group has only 9 applicants.

RETRIEVAL & GENERATION

# Clauses and the draft

## Policy Corpus

**Project paraphrases:** Eight EBA/GL/2020/06 clauses indexed once.

**Official guideline:** Full 89-page report grounds search.

## Draft Guardrails

**Citations & review:** Flags unretrieved IDs or missing “manual review.”

**Prohibited claims:** Warns on loan denial, “grounds for denial,” or invented missed payments.

**Completeness:** Flags any sentence that stops halfway.

# Live demo

INTERPRETABILITY

# How to read the explanation

**Baseline:** Starts at 28% average default risk.

**Red bars:** Push predicted risk upward.

**Teal bars:** Lower predicted risk downward.

**Dual views:** Top bar shows percent; log-odds show additive math.

## Observed risk indicators

Only two observed indicators qualify: checking below 0 DM and past payment delays.

**A positive SHAP value is not itself an adverse reason.**

INTERPRETABILITY · VALIDATION

# Do the explanations hold up?

**$\sim 10^{-6}$**

Fidelity: max gap between baseline + SHAP and the model's log-odds (4 demo applicants)

**0.92**

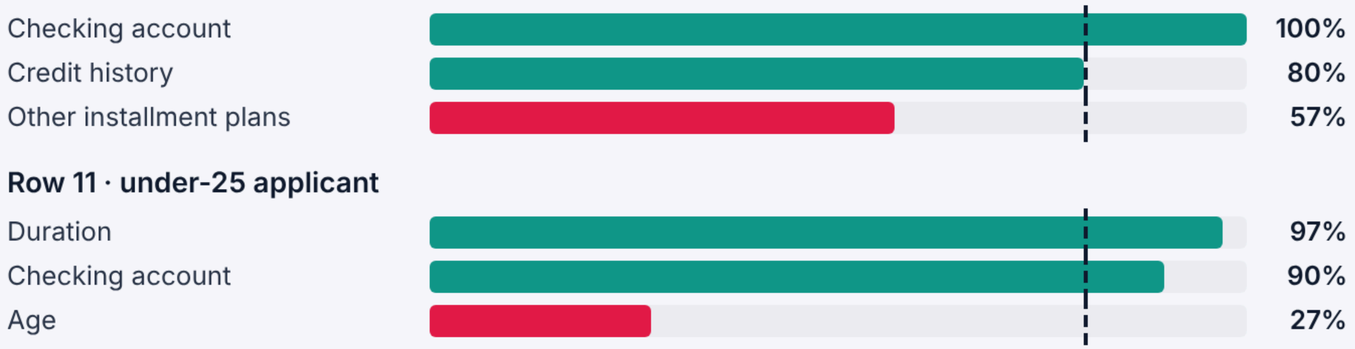
Global importance ranking vs. saved model, median Spearman over 30 bootstrap retrains

**38–98%**

Row 740 predicted risk across retrains (saved model: 83%). Flagged higher risk in 90% of runs

**Share of 30 retrains keeping the factor in the top 3**

Dashed line: 80% stability threshold (project choice, not a regulatory value)




**Top two factors are stable**

Matches the two focus factors sent to retrieval. Same top 3 under two interventional SHAP backgrounds for 3 of 4 applicants.

**The third factor is not**

Rank 3 drops out in most retrains. It should be shown as context, never cited as a reason.

**The score itself moves**

Retraining on resampled data shifts row 740 by tens of points. Supports a person reviewing every case.

GOVERNANCE

# Limits and choices

### 100% Human Referral

No calibrated auto-decision band or rejection cutoff.

### Small Sample Size

1,000 rows total; 32 under-25 test applicants.

### Draft Hallucination Risk

Automated warnings flag invented facts for reviewers.

### Deterministic Output

Fixed model file guarantees identical SHAP explanations.

# Thank you for your attention.
# Questions?